

TOTHBOT V2 — SYSTEM ARCHITECTURE OVERVIEW

The Complete System — All Architectural Decisions + All Flow Diagrams

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This document captures all architectural decisions, FP/DP analyses, and system specifications from the TothBot Kraken architecture sessions of March 2026. Includes all six system flow diagrams showing connections between Kraken, TothBot, and CIATS as one unified organism.

TOTHBOT — KRAKEN SYSTEM ARCHITECTURE OVERVIEW

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PURPOSE: This document captures all architectural decisions, FP/DP analyses, and system specifications from the TothBot Kraken architecture sessions of March 2026. Includes all six system flow diagrams showing connections between Kraken, TothBot, and CIATS as one unified organism.

Diagrams in this document (dv1_9 — Diagrams 1 and 6 updated):

Diagram 1 — Kraken API Connection Architecture

(dv1_6: amend_order primary, batch_cancel added, WS ping A-7, zombie A-8, balances seq A-9, per-pair status, L3 URL corrected) (dv1_7: executions seq gap detection A-10, GetOpenOrders gap recovery)
(dv1_9: GAP-A order_status:true CRITICAL FIX, GAP-C price_increment, GAP-D qty_min/cost_min, GAP-E max_size=10MB, IMP-11 maxratecount, IMP-13 warnings[])

Diagram 2 — TothBot Trading Pipeline

(dv1_6: per-pair instrument status check added before Gate 1)

Diagram 3 — Exit Architecture: Ground Zero Redesign (dv1_6: title updated)

Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging (dv1_6: title updated)

Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle (dv1_6: title updated)

Diagram 6 — Complete System: One CIATS-Owned Organism (dv1_9: GAP-A through F, IMP-11 through 17)

1. System Identity and Mission

1.1 What TothBot Is

TothBot is an automated cryptocurrency spot trading system on Kraken exchange. It runs 24/7/365 on a Hetzner VPS, processes 5-minute OHLC candle close events as its system clock, and executes long-only trades using a momentum-based signal engine.

TothBot is not an external system connected to Kraken — it IS Kraken's event processor. Every action is a response to a Kraken push event. No timers. No polling. Idle and silent until Kraken speaks.

1.2 The One Organism Principle

Three functional zones, one organism:

Kraken: generates events, executes orders, holds resting TP limit orders and

off-book emergency SL stop orders on matching engine

TothBot: processes Kraken events, executes trading strategy, manages positions

CIATS: learns from event records, continuously improves processing rules

PRIMARY MISSION: Maximize profit. Minimize loss. With an emphasis on minimizing loss.

Loss prevention takes priority over profit generation. Everything else is secondary.

1.3 Operating Environment

Exchange: Kraken Spot (Long Module — Short deferred)

Language: Python 3.12.3

VPS: Hetzner CPX22 — 87.99.141.44 — Ashburn VA

(Kraken Spot servers: AWS us-west-2 Oregon) ★ Evaluate Hillsboro OR [dv1_6 IMP-01 — adjacent to Kraken AWS]

System Clock: Kraken OHLC 5-minute candle close event (WS v2 push)

Fee Baseline: Maker 0.16% / Taker 0.26% (CIATS monitors actuals)

Operation: 24/7/365 — CIATS may add stand-down periods via PDCA if data supports

Starting Portfolio: \$1,000 Long Module

Sacred constraint: Net 1:1.5 R:R — HARDCODED — the only number not owned by CIATS

2. Kraken API Connection Architecture

All connection points verified across 3 independent analysis passes (15 sources) plus

2 deep validation sessions (144+ sources total) plus 10-search FP/DP Kraken validation

[dv1_6] (126 sources).

2.1 Key Connection Points

Public WS v2:

instrument — asset specs + per-pair status cache ★ dv1_6 price_increment / qty_min / cost_min ★ GAP-C/D
dv1_9 status — global engine state (online/post_only/cancel_only/maintenance) ohlc(5m) — SYSTEM CLOCK
(pipeline trigger, fires on trade events) ohlc(60) — 1H HTF cache for Gate 4 ★ NEW dv1_5 ticker — price
data, event_trigger configurable (trades or bbo)

Private WS v2 inbound:

executions — fills/expiration/SL-TP triggers, ratecounter:true (A-1), snap_orders:true,
snap_trades:true, order_status:true on reconnect (A-3), sequence gap detection ★ A-10 dv1_7 CRITICAL ★
GAP-A dv1_9: order_status:true MANDATORY — without it amended and restated exec_types are silently
dropped, breaking A-12 and A-13 balances — real-time settlement, sequence gap detection ★ A-9 dv1_6

Private WS v2 outbound:

add_order — entry: limit/post_only/GTD/cl_ord_id/stp_type:cancel-newest amend_order — PRIMARY amendment
method (edit_order DEPRECATED ★ dv1_6) batch_add — SL+TP atomic on fill cancel_order — exit leg
cleanup, req_id tracked batch_cancel — Full Halt multi-order cancel ★ dv1_6 All messages carry req_id
(A-2), cl_ord_id (A-2), stp_type:cancel-newest (A-4)

REST:

GetWebSocketsToken — startup/reconnect GetOHLCData — daily regime computation 00:00 UTC ★ NEW dv1_5
CRITICAL: exclude response[-1] always ★ dv1_6 GetOpenOrders — restart reconcile fallback / executions
gap recovery ★ dv1_7 GetAccountBalance — restart reconcile fallback / balances gap recovery ★ dv1_6
CancelAllOrders — kill switch emergency only

Matching Engine:

Resting TP limit orders and off-book emergency SL stop orders survive any TothBot failure independently
— crash protection always active

EVALUATED AND REJECTED:

cancel_all_orders_after (WS v2 DMS) — destroys crash protection (AR-001) FIX API cancel-on-disconnect —
same architectural conflict (AR-001) OTO conditional orders — uses limit price not fill price, violates
R:R (AR-008)

2.2 Four Kraken Engine States

online — All orders. Signal pipeline runs normally.

post_only — TothBot limit+post_only entries ARE valid. Trading continues.

cancel_only — No new entries. Exit Controller can cancel. Resting orders active.

maintenance — No orders, no cancels. Resting TP and emergSL active on engine.

2.3 API Rate Limits — Intermediate Tier (C-2)

Rate limits shared across REST, WebSockets, and FIX — no separate per-protocol budget.

Rate counter threshold (per pair): 125

Rate counter decay (per second): 2.34

Max open orders (per pair): 80

Add order fixed cost: +1

Amend order cost ★ dv1_6 (PRIMARY): +1 fixed, +3(<5s), +2(<10s), +1(<15s)

Edit order (DEPRECATED ★ dv1_6): +1 fixed, +6(<5s), +5(<10s) — DO NOT USE

Batch add fixed cost: +(n/2)

Cancel order cost (< 5s old): +8

Cancel order cost (< 10s old): +6

Cancel order cost (< 300s old): +1

Batch cancel cost ★ dv1_6: Cumulative sum of individual order cancel costs

amend_order is 43% cheaper than edit_order for orders resting <5s (+4 vs +7).

amend_order preserves queue position (unlike cancel+resubmit).

2.4 WS Operational Requirements (A-1 through A-10)

A-1 Rate counter monitoring via executions channel: executions subscription accepts ratecounter:true. Real-time rate counter per pair included at zero additional connection cost. CIATS EWMA Monitor adds rate_counter_by_pair as third monitoring signal. Assign to 0511003 and 0211001.

A-2 req_id and cl_ord_id on all outbound WS messages: All outbound messages carry req_id (client-specified integer). All add_order and batch_add additionally carry cl_ord_id (up to 18 characters). TothBot assigns cl_ord_id at dispatch time before Kraken response. Position Mirror records cl_ord_id immediately at dispatch — no async gap. WS Manager maintains pending-request registry. Assign to 0511001, 0511003, 0511005.

A-3 Round-trip latency monitoring and WS reconciliation: All WS v2 responses include time_in and time_out to nanosecond precision. Logger records these fields. CIATS EWMA Monitor adds ws_roundtrip_latency_ms. executions channel supports snap_orders:true on subscription — preferred over REST GetOpenOrders for restart reconciliation. Assign to 0511001, 0511006, 0211001.

A-4 stp_type:cancel-newest on all add_order calls: All add_order and batch_add include stp_type:cancel-newest. Self Trade Prevention prevents TothBot from inadvertently filling against its own resting orders. Zero-cost prevention. Assign to 0511003.

A-5 Reconnect behavior distinction: (1) Random drop: reconnect instantly, up to a handful of attempts. (2) After confirmed trading engine maintenance: minimum 5-second delay. Both implemented as separate paths in WS Manager. Assign to 0511001.

A-6 (dv1_5) OTO conditional orders — REJECTED: OTO uses limit price not actual fill price — violates sacred R:R. Creates Position Mirror race condition. batch_add on fill confirmed correct. See AR-008.

A-7 (dv1_6) ★ WS Ping / Keepalive Strategy: Kraken WS server disconnects after ~60 seconds of inactivity. Cloudflare idle timeout ~100 seconds. WS Manager MUST send {"method":"ping"} on each active connection every 30 seconds. Expects {"method":"pong"} within 10 seconds. No pong within 10 seconds = dead connection. Reconnect immediately. Application-level ping, distinct from TCP keepalive. Assign to 0511001, 1011002.

A-8 (dv1_6) ★ Zombie Connection Detection: A WS connection can appear alive (ping/pong passing) while delivering no real market data. WS Manager MUST track last_real_data_time per connection using time.monotonic(). Only actual market data events reset this timestamp — heartbeat messages alone do NOT. If elapsed > 90 seconds: log ZOMBIE_CONNECTION_DETECTED, alert operator, reconnect. Assign to 0511001, 1011002.

A-9 (dv1_6) ★ Balances Channel Sequence Gap Detection: Every balances channel message carries a sequence integer. WS Manager MUST track last-seen sequence. Any gap (skip > 1) indicates a missed ledger event. On gap: alert immediately, trigger REST GetAccountBalance reconciliation. Missed balance events cause incorrect available capital calculation — direct loss risk. Assign to 0511001, 1011002.

A-10 (dv1_7) ★ Executions Channel Sequence Gap Detection: Every executions channel message carries a sequence integer. WS Manager MUST track last-seen sequence independently of balances. Any gap (skip > 1) indicates missed order/fill events. On gap: alert immediately, trigger REST GetOpenOrders reconciliation to restore Position Mirror accuracy. Missed execution events mean TothBot does not know a position is open or closed — direct loss risk. Higher severity than A-9 (balances): a missed fill leaves Position Mirror in stale state; the next pipeline evaluation operates on incorrect position data. Sequence tracking resets on each new subscription (reconnect). Do not gap-detect across sessions. Assign to 0511001, 1011002.

A-11 (dv1_8) ★ Executions Channel: cancel_reason Deprecated — Use reason Field: The cancel_reason field in executions channel events is DEPRECATED since WS v2.0.0 (December 2023). All code parsing cancellation events MUST read msg["reason"], not msg["cancel_reason"]. cancel_reason returns None on all modern messages. Silent reason loss corrupts CIATS EWMA rejection_rate_by_regime monitoring accuracy. Assign to 1011002 WM-EXE-002 through WM-EXE-004.

A-12 (dv1_8) ★ Executions Channel: Complete exec_type Enum Handling Required: WS Manager MUST handle all exec_type values: pending_new, new, trade, filled, iceberg_refill, canceled, expired, amended, restated, status. Using a partial dispatch table causes unknown exec_type events to silently fall through. Use handle_exec_unknown for unrecognized values — log full payload, never silently ignore. TothBot primary events: trade (trigger batch_add), filled, expired (log GTD expiry), canceled (log with reason), amended and restated: log only, no Position Mirror change. Assign to 1011002 WM-EXE-005 through WM-EXE-007.

A-13 (dv1_8) ★ Executions Channel: restated exec_type Must Not Trigger Position Change: restated fires when Kraken's matching engine amends an order for maintenance purposes. It is NOT a fill, NOT a cancel, NOT a user action. On restated: (1) log with reason field, (2) alert operator (informational), (3) do NOT update Position Mirror, (4) if the restated order is a resting TP or emergSL, escalate to HIGH severity alert and queue a reconciliation check on the next pipeline cycle. The restated event on a resting crash-protection order indicates Kraken modified our stop — must be visible. Assign to 1011002 WM-EXE-008 through WM-EXE-010.

A-14 (dv1_8) ★ Spot REST Authentication: Confirmed Unchanged — PL-014 RESOLVED: The October 1, 2025 authentication breaking change applied to Futures REST ONLY. Kraken Spot REST authentication is UNCHANGED: HMAC-SHA512 of (URI path + SHA256 (nonce + POST data)) using base64-decoded secret. No code changes required. PL-014 is fully resolved. No Bill verification action required. Assign to 0411001 WS-AUTH-001 through WS-AUTH-003.

A-15 (dv1_9) ★ Executions Channel: order_status:true MANDATORY — GAP-A CRITICAL FIX: Without order_status:true in the executions subscription, the channel only delivers: new, filled, canceled, expired. The exec_types amended and restated are SILENTLY DROPPED. This means: (1) A-13 restated protection for crash-protection orders is completely non-functional without this flag — the restated event never arrives. (2) A-12 dispatch table handles 10 exec_types but receives only 4. The subscription is already specified correctly (WS-EXE-001 in 0411001) but this architectural consequence must be explicitly documented. Assign to 0411001 WS-OSTA-001 through WS-OSTA-003, 1011002 WM-EXE-011/-012.

A-16 (dv1_9) ★ OHLC Channel: interval_begin Replaces Deprecated timestamp — GAP-B: The Kraken WS v2 OHLC channel deprecated the timestamp field. Subscription ACK warns: "timestamp is deprecated, use interval_begin". TothBot MUST parse interval_begin (RFC3339 nanosecond string) as the candle start time, not timestamp. Assign to 0411001 WS-OHLC-002 through -005, 1011002 WM-OHLC-001 through -003.

A-17 (dv1_9) ★ Instrument Channel: price_increment / qty_min / cost_min — GAP-C/D: (GAP-C) The instrument channel deprecated tick_size. Use price_increment exclusively. Subscription ACK warns: "tick_size is deprecated, use price_increment". (GAP-D) The instrument channel provides qty_min (minimum order quantity) and cost_min (minimum order cost = price × qty). Gate 8 Position Sizer MUST validate both before submitting any add_order. Violation causes immediate order rejection. Assign to 0411001 WS-INST-001 through -007, 1011002 WM-INST-001 through -005.

A-18 (dv1_9) ★ WebSocket Library max_size = 10MB — GAP-E: Python websockets library default max_size = 1MB. The Kraken instrument channel snapshot for 500+ pairs can exceed 1MB, causing PayloadTooBig and silent disconnect. All WS connect() calls MUST set max_size=10*1024*1024 (10MB) and open_timeout=10. Assign to 0411001 WS-LIB-001 through -003, 1011002 WM-LIB-001 through -003.

A-19 (dv1_9) ★ maxratecount from Executions ACK is Operative Rate Threshold — IMP-11: The executions subscription ACK returns maxratecount — the actual operative rate counter ceiling for this user tier. TothBot MUST extract and use maxratecount as the operative threshold for all rate counter monitoring, not the hardcoded value 125. If tier changes, maxratecount automatically updates on next reconnect. Assign to 0411001 WS-RL-003, WS-EXEC-001a, 1011002 WM-RATE-001 through -003.

2.5 Data Architecture: Pre-Computation Cache (NEW dv1_5 — I-2)

HOT PATH RULE: The 5-minute pipeline hot path reads cached values ONLY. No computation occurs during pipeline evaluation. All indicators maintained as incremental running calculations updated on each event trigger.

IMP-06 (dv1_8) ★ aiohttp Mandatory for All REST Calls: All Kraken REST calls MUST use aiohttp (async HTTP). Never use the requests library inside asyncio — it blocks the event loop during the entire call duration. Create one aiohttp.ClientSession at TothBot startup, reuse for all REST calls, close at shutdown. Assign to 1011002 WM-REST-001 through WM-REST-005.

IMP-07 (dv1_8) ★ asyncio TaskManager Pattern — Mandatory for 24/7 Operation: All asyncio.create_task() calls MUST store a strong reference AND register a done callback via task.add_done_callback(task_set.discard). Failure to do so causes memory leaks in long-running processes: the task set grows indefinitely until OOM. All task creation routed through a central create_task() helper function. Long-running tasks (ping, zombie, watchdog) stored as named instance variables, cancelled and awaited explicitly during reconnect cleanup. Assign to 1011002 WM-TASK-001 through WM-TASK-004.

IMP-08 (dv1_8) ★ systemd Watchdog Integration — Frozen Process Detection: TothBot MUST run as a systemd service with WatchdogSec=120 and Type=notify. A separate asyncio.Task pings systemd sd_notify every 30 seconds via WATCHDOG=1 message. If the event loop hangs (frozen process), the watchdog stops, systemd detects no ping within 120s, and restarts TothBot. Without this, a frozen-but-alive TothBot with open positions is undetected indefinitely. Assign to 1011013 VPS_Deployment_Coding_Spec VD-WD-001 through VD-WD-006.

Value Pre-computed by Update Trigger Cache Read On

Regime classification Regime Engine Daily 00:00 UTC REST GetOHLCDData Every 5m eval (Gate 3+6)

[CRITICAL: exclude response[-1] ★]

1H EMA(20) and EMA(50) Signal Pipeline 1H candle close ohlc(60) WS event Gate 4

(parse interval_begin ★ A-16 dv1_9)

ATR(14) per pair Signal Pipeline Each 5m candle close (incremental) Gate 8 + Exit Controller

24h liquidity per pair (USD) Signal Pipeline Daily ticker or daily OHLC pull Gate 2

Per-pair instrument status ★ WS Manager instrument channel events (live) Gates 1/2 pre-filter

3. TothBot Trading Pipeline

The pipeline fires on every 5-minute OHLC candle close event. Synchronous and event-driven.

8 gates. Gate 3 (Regime Pre-Filter) new in dv1_5. Per-pair status check new in dv1_6.

NOTE — OHLC fires on trade events, not wall clock: If no trades occur after an interval border, no message sent until next trade. Liquidity Gate (\$500k/day) reduces this risk.

Same stall behavior applies to ohlc(60). TothBot must handle stalled system clock.

Addressed in 0511002 Signal Pipeline Specification.

3.1 Gate Summary

PRE-STEP: Parameter Store Snapshot Frozen (NEW dv1_5 — I-4)

CIATS params captured as read-only snapshot at pipeline start. Consistent throughout full eval. CIATS writes applied only at inter-trade boundaries.

PRE-STEP: Pre-Computation Cache Read (NEW dv1_5 — I-2)

Hot path reads: regime cache | 1H EMA cache (ohlc(60)) | ATR(14) running value | 24h liquidity | per-pair status. Zero computation on hot path.

★ dv1_6: Per-pair Instrument Status Check

Skip pairs in reduce_only / work_in_progress / delisted / limit_only state. Only online or post_only pairs proceed into pipeline. Prevents wasted rate counter.

Gate 1: State Machine

Kraken engine = maintenance or cancel_only: WAIT

Gate 2: Liquidity Gate ★

24h volume < \$500k USD: SKIP → LIQUIDITY_REJECTED

Gate 3: Regime Pre-Filter ★ NEW dv1_5 (I-1)

TRENDING_NEG (any vol) or NON_DIR+ELEVATED: STOP → REGIME_BLOCKED Eliminates SSS computation on all no-trade regimes.

Gate 4: 1H HTF Confirmation ★ (was Gate 3)

1H EMA(20) <= EMA(50) for TRENDING_POS entries: SKIP → HTF_GATE_REJECTED Cache read from ohlc(60) WS subscription (NEW dv1_5).

SSS Signal Engine

Scores 5-min candle. Only reached if Gate 3 pre-filter passed.

Gate 5: Selection Controller (was Gate 4)

Signal fails any of 6 quality gates: SKIP → SELECTION_REJECTED

Gate 6: Regime Gate ★ (was Gate 5)

NON_DIR+NORMAL: 50% position size. All NO ENTRY cases blocked by Gate 3.

Gate 7: Risk Guard (was Gate 6)

Drawdown halt / concentration / exposure. Pass or HALT. ★ Ticker-driven continuous drawdown monitoring between candle closes (I-7 dv1_5)

Gate 8: Position Sizer ★ (was Gate 7)

MAE = $ATR(14) \times 1.5x$ (incremental), emergSL = $ATR(14) \times 3.0x$, TP = R:R formula. Never blocks — only sizes.

EVIDENCE BASE: 15-20% higher win rates with daily+hourly confirmation (5 independent sources 2024-2025). Market Cipher: accuracy drops 65% in TRENDING to 45-50% in NON_DIRECTIONAL. Break-even 40% for 1:1.5 R:R.

4. Exit Architecture — Ground Zero Redesign for Kraken

Rebuilt from zero for Kraken. Exit Controller owns all normal exits. Kraken off-book emergency SL stop order is crash protection only. OTO evaluated and rejected (AR-008).

4.1 Three-Layer Summary

Layer 1a — Signal-Based Exit:

Signal decay / time expiry / momentum loss. WS cancel TP + market sell.

Layer 1b — TP Fill:

Resting TP limit on Kraken matching engine fills. EC cancels emergSL after fill confirm.

Layer 2 ★ — MAE Threshold:

price - entry $\geq ATR(14) \times 1.5x$. SEQUENCE CRITICAL: (1) cancel TP (2) cancel emergSL THEN (3) market sell. CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6): req_id registry tracks each cancel ACK. On timeout: check executions channel state. Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT + hold position + alert operator. NEVER issue market sell with ambiguous order state. Assign to 0511001 and 0511004. MPP RISK (C-1 dv1_4): market sell may reject on wide spread. EC MUST handle MPP rejection and retry. Assign to 0511004.

Layer 3 ★ — Emergency Brake:

VPS/TothBot/internet failure ONLY. Off-book emergSL stop order at $ATR(14) \times 3.0x$ on Kraken matching engine. NEVER in normal operation. MPP RISK (C-1 dv1_4): triggered market order subject to MPP on wide spread. Assign to 0411001.

4.2 R:R Formula

mae_pct = $ATR(14) \times 1.5x / \text{entry_fill_price}$ (CIATS-owned, uses ACTUAL fill price)

net_loss = mae_pct + 0.16% + 0.26% (entry maker + MAE taker exit)

net_gain = net_loss $\times 1.5$ *** SACRED — hardcoded ***

gross_target = net_gain + 0.16% (TP maker exit)

emergSL_dist = $ATR(14) \times 3.0x / \text{entry_fill_price}$ (NEVER in R:R formula)

NOTE: All calculations use actual fill price, not limit price. OTO uses limit price — this is why OTO was rejected (AR-008).

5. Regime Taxonomy

Six-regime grid is foundation of CIATS regime-segmented analysis. Every trade tagged with both asset_regime and market_regime (BTC proxy). Computed daily at 00:00 UTC.

5.1 Computation

DATA SOURCE (NEW dv1_5 — FLAG 2 — FP/DP VERIFIED):

Daily candles via REST GetOHLCData called once per monitored pair at 00:00 UTC.

Returns 720 entries.

CRITICAL dv1_6 (GAP-03): The last entry (response[-1]) is ALWAYS the current not-yet-committed candle. MUST be excluded from ALL calculations (ADX, EMA, ATR).

Use only response[:1]. Including the uncommitted candle corrupts all regime indicators and produces incorrect regime classification. See 1011012 Regime_Engine_Coding_Spec.

Directional state: ADX(14) + EMA(20) vs EMA(50) on DAILY candles (response[:1] ONLY)

Volatility state: ATR(14) percentile rank in 50-day rolling window

TRENDING_POSITIVE: ADX(14) > 25 AND EMA(20) > EMA(50)

TRENDING_NEGATIVE: ADX(14) > 25 AND EMA(20) < EMA(50)

NON_DIRECTIONAL: ADX(14) <= 25

NORMAL_VOL: ATR percentile <= 67th

ELEVATED_VOL: ATR percentile > 67th

asset_regime + market_regime (BTC/USD proxy) both tagged on every trade

ALL REGIME THRESHOLDS ARE CIATS-OWNED: ADX threshold 25, ATR percentile 67th, EMA periods 20/50, rolling window 50 days — all starting values. CIATS validates each after 600 closed trades (100+ per regime bucket).

6. CIATS Architecture and Data Flows

CIATS is fully asynchronous. Never blocks the trading critical path. Reads only from the Logger. Writes only to the Parameter Store. Deming PDCA governs every change.

6.1 Two Data Streams

Stream 1: EWMA Monitor

288 candle evals/day. Valid at 50 evals (~4 hrs). $\lambda=0.2$ [FP/DP validated dv1_6: half-life=3.1 evals]. Three monitoring signals: (1) rejection_rate_by_regime (F6) (2) rate_counter_by_pair from executions channel (A-1) (3) ws_roundtrip_latency_ms from WS response timestamps (A-3) MONITORING SIGNALS ONLY — never drives parameter changes.

Stream 2: Trade Outcome Bus

Fires on each position close. Captures: symbol, entry/exit price, hold duration, MAE reached, fees, exit reason, both regime tags, signal params at entry, actual R:R, P/L.

LOGGER — ASYNC BOUNDED QUEUE (NEW dv1_5 — I-5):

Logger writes go to an in-memory bounded queue. Queue size: 10,000 entries

[FP/DP validated dv1_6: 20 positions x 10 events/candle x 50-candle buffer].

Separate async writer flushes to disk. Logger NEVER blocks the trading hot path.

Queue must be bounded with immediate alert if queue fills. Assign to 0511006.

PARAMETER STORE — SNAPSHOT PATTERN (NEW dv1_5 — I-4):

TothBot takes frozen snapshot of all CIATS parameters at start of each pipeline eval.

Pipeline uses snapshot throughout. CIATS writes queued and applied only at confirmed inter-trade boundaries. Prevents parameter inconsistency within single pipeline eval.

Assign to 0511002 and 0211001.

6.2 Statistical Thresholds

50 candle evals: EWMA Monitor valid. Monitoring only. $\lambda=0.2$ validated.

200 trades HARD FLOOR: Inference valid. Proposal Engine activates. Half-Kelly activates.

$SE(win_rate) \sim 3.5\%$, $SE(Sharpe) \sim 7.2\%$ [FP/DP validated dv1_6].

600 trades: Per-regime analysis. 100+ per bucket. MW power ~85% at $\alpha=0.01$.

Rolling 200-300: Current performance window.

Full corpus: Zero-cost historical library.

Mann-Whitney alpha (dv1_6) ★: $\alpha=0.01$ (one-sided). 99% confidence required.

Spearman rho threshold (dv1_6) ★: $|\rho| > 0.3$ AND $p < 0.05$ (both conditions).

CUSUM parameters (dv1_6) ★: $k=0.5 \cdot \sigma$, $h=4 \cdot \sigma$. Starting values.

PDCA minimum interval (dv1_6) ★: Min 50 trades between any two parameter changes.

7. Complete System: One CIATS-Owned Organism

CIATS OWNS EVERYTHING EXCEPT 1:1.5 R:R: All parameters, thresholds, sizing decisions, halt thresholds, multipliers, and trading rules are CIATS-owned starting values.

8. Locked Architectural Decisions

AR-001: No Dead Man Switch — crash protection is resting TP + emergSL on matching engine. DMS and FIX cancel-on-disconnect both REJECTED (destroy crash protection). AR-002: Position Mirror — symbol-keyed dict, O(1) lookup. (NEW dv1_5 — I-3) AR-003: Restart reconciliation — PREFERRED: snap_orders:true. FALLBACK: REST. AR-004: Two API key pairs — DATA + TRADE. Nonce window 10,000ms. AR-005: WS v2 only. All future Kraken enhancements in v2 only. AR-006: Entry order — limit/post_only/GTD/cl_ord_id/stp_type:cancel-newest. AR-007: TP placement — resting limit via batch_add on fill. Actual fill price for R:R. AR-008: Emergency SL — stop-market via batch_add atomic with TP. ATR(14) x 3.0x. OTO evaluated and REJECTED: uses limit price, violates R:R, Position Mirror race. AR-009: Short deferred. AR-010: 24/7/365. CIATS adds stand-down via PDCA if data supports. AR-011: Net 1:1.5 R:R — SACRED — HARDCODED — NOT CIATS-OWNED. AR-012: Fee baseline — Maker 0.16% / Taker 0.26%. AR-013: Kraken Pro Intermediate tier confirmed. AR-014: Logger async bounded queue (10,000 entries). Never blocks hot path. (dv1_5 I-5) AR-015: WS dispatch table O(1) routing on channel+message type. (dv1_5 I-8) AR-016: ATR(14) incremental running calculation. O(1) per update. (dv1_5 I-9) AR-017: Daily regime via REST GetOHLCData at 00:00 UTC. CRITICAL: exclude response[-1]. (dv1_5 FLAG 2) AR-018 (dv1_6) ★: amend_order PRIMARY. edit_order DEPRECATED. AR-019 (dv1_6) ★: batch_cancel for Full Halt multi-order cancellation. AR-020 (dv1_6) ★: Per-pair instrument status cache from instrument channel. AR-021 (dv1_7) ★: Executions channel sequence gap detection → REST GetOpenOrders reconciliation. Missed executions = Position Mirror desync = loss risk. AR-022 (dv1_8) ★: cancel_reason field deprecated — use reason field in all executions channel parsing. cancel_reason returns None silently on current messages. AR-023 (dv1_8) ★: Complete exec_type dispatch table required. All 10 exec_type values handled explicitly. Unknown values logged, never silently dropped. AR-024 (dv1_8) ★: restated exec_type = engine maintenance amend, NOT fill/cancel. Must not trigger Position Mirror update. Elevated alert if resting order. AR-025 (dv1_8) ★: All REST calls via aiohttp (IMP-06). asyncio TaskManager pattern mandatory for 24/7 operation (IMP-07). systemd WatchdogSec=120 with 30-second asyncio ping loop (IMP-08). See 1011013 VPS_Deployment_Spec. AR-026 (dv1_9) ★: order_status:true MANDATORY in executions subscription. Without it amended and restated exec_types are silently dropped — A-12/A-13 blind. AR-027 (dv1_9) ★: interval_begin replaces deprecated timestamp in OHLC WS v2 channel. AR-028 (dv1_9) ★: price_increment replaces deprecated tick_size in instrument channel. qty_min and cost_min validated in Gate 8 before every add_order. AR-029 (dv1_9) ★: WS connect max_size=10MB + open_timeout=10 on all connections. AR-030 (dv1_9) ★: maxratecount from executions subscription ACK is the operative rate counter ceiling. Use it, not the hardcoded 125. AR-031 (dv1_9) ★: Subscription ACK warnings[] parsed and logged on every subscribe call. QueryOrdersInfo REST endpoint for targeted single-order reconciliation.

9. Complete CIATS Parameter Registry

ALL PARAMETERS ARE CIATS-OWNED STARTING VALUES.

tradeable_pct: 50% of portfolio
per_trade_pct: 5% → Half-Kelly at 200 trades (update every 50 thereafter)
max_concurrent: 20 positions
mae_mult (ATR x): 1.5x ATR(14)
emergency_sl_mult: 3.0x ATR(14)
entry_timeout_sec: 30-60s GTD
full_halt_drawdown: 10% of portfolio
session_pause_drawdown: 5% of portfolio
signal_thresholds: SSS gates (derived)
adx_threshold: 25
atr_percentile_threshold: 67th
htf_ema_periods: 20/50 daily
min_volume_usd_daily: \$500k USD

1:1.5 R:R net: HARDCODED — SACRED — NOT CIATS-OWNED

10. Six FP/DP Validated System Improvements

Derived from 80+ sources across 3 independent research passes plus 2 deep validation sessions (144+ sources) plus 10-search Kraken validation [dv1_6] (126 sources).
5+ sources per parameter.

Finding 1: 1H HTF Confirmation Gate [TothBot]

Gap: No layer between daily regime and 5-min entry.

Evidence: 15-20% higher win rates with daily+hourly filter (5 sources 2024-2025).

Change: 1H EMA(20) > EMA(50) required for TRENDING_POSITIVE. Cache from ohlc(60).

Effect: Removes false signals during 1H pullbacks. Zero API cost.

Finding 2: ATR Dynamic Stop Distances [TothBot/Exit]

Gap: Fixed 2% stop ignores real-time market volatility.

Evidence: +15% performance, -22% max drawdown vs fixed stops (LuxAlgo + 4 sources).

Change: MAE ATR(14) x 1.5x. Emergency SL ATR(14) x 3.0x. Incremental (I-9).

Effect: Stop adapts to conditions. R:R formula preserved.

Finding 3: Regime Gate [TothBot]

Gap: Long momentum system trading in absent-edge conditions.

Evidence: Accuracy 65%→45-50% in NON_DIRECTIONAL (Market Cipher). Break-even 40%.

Change: NON_DIR+ELEV and TRENDING_NEG: no entry (Gate 3). NON_DIR+NORM: 50% (Gate 6).

Daily candles via REST GetOHLCData. CRITICAL: exclude response[-1] [dv1_6].

Effect: Stops trading when documented edge is absent. Pure loss minimization.

Finding 4: Minimum Liquidity Gate [TothBot]

Gap: No minimum 24h volume on asset selection.

Evidence: \$1M+/day minimum for clean execution (Coin Bureau + 4 sources 2024-2025).

Change: min_24h_volume_usd \$500k starting. Checked daily via ticker.

Effect: Protects fee formula. Ensures clean post_only fills and SL execution.

Finding 5: CIATS Half-Kelly Pathway [CIATS]

Gap: Fixed 5% sizing had no defined data-driven transition.

Evidence: Half-Kelly: 75% optimal growth at 50% less drawdown (MacLean-Thorp-Ziemba + 4).

FP/DP validated [dv1_6].

Change: At 200 trades: $K\% = W - (1-W)/R$, apply 50%. Update every 50 trades. No ceilings.

Effect: Data-driven sizing replaces arbitrary fixed fraction.

Finding 6: post_only Rejection Monitoring [CIATS]

Gap: Rejection events logged but unused by CIATS EWMA.

Evidence: Rejection rate = direct signal of entry placement quality (Cube Exchange + 4).

Change: CIATS EWMA adds rejection_rate_by_regime stream.

Effect: Free diagnostic signal. Entry optimization pathway activated.

11. Complete Trade Lifecycle — Narrative Reference

11.1 Normal Trade (Entry to TP or Exit Controller)

OHLC 5-min candle close fires on public WS v2

Parameter Store snapshot frozen — all CIATS params captured (NEW dv1_5 — I-4)

Pre-computation cache read — regime, 1H EMA, ATR(14) incremental, 24h liquidity (I-2)

★ Per-pair instrument status check — skip reduce_only/work_in_progress/delisted [dv1_6]

Gate 1: State Machine — online or post_only: continue; else halt and wait

Gate 2: Liquidity Gate — 24h volume >= \$500k: continue; else skip

Gate 3: Regime Pre-Filter — TRENDING_NEG or NON_DIR+ELEV: STOP; else continue (I-1)

Gate 4: 1H HTF Gate — 1H EMA(20) > EMA(50) for TRENDING_POS: continue; else skip

Signal Engine: SSS score computed on 5-min candle

Gate 5: Selection Controller — 6-gate quality evaluation: pass or reject

Gate 6: Regime Gate — TRENDING_POS: proceed; NON_DIR+NORM: 50% size

Gate 7: Risk Guard — drawdown/concentration/exposure checks: pass or halt

Gate 8: Position Sizer — $MAE=ATR \times 1.5x$, $emergSL=ATR \times 3.0x$, $TP=R:R$ formula

Execution: WS add_order (limit/post_only/GTD 30-60s/cl_ord_id/stp_type:cancel-newest)

If expired/rejected: log reason code → CIATS EWMA monitors rejection rate. Done.

If filled: WS batch_add → TP (resting limit) + emergSL (stop-market) atomic

Position Mirror records: symbol (key), entry_fill_price, qty, TP id, cl_ord_id, emergSL id

Exit Controller monitors via ticker (continuous drawdown check — I-7)

Logger records full entry via async queue. CIATS Trade Outcome Bus ready.

TP fills: EC cancels emergSL. Or EC fires Layer 1/2 exit.

Logger records full exit. CIATS Trade Outcome Bus fires with complete trade record.

11.2 Layer 2 MAE Exit (Adverse Price)

Exit Controller detects: $current_price - entry \geq ATR(14) \times 1.5x$

WS cancel_order: cancel TP resting order

WS cancel_order: cancel emergSL stop order (SEQUENCE CRITICAL: both cancels before sell)

CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6):

req_id registry tracks each cancel ACK. On timeout: check executions channel state. Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT and hold position. NEVER issue market sell with ambiguous order state. Assign to 0511001 and 0511004.

WS add_order: market sell (taker 0.26%)

MPP NOTE (C-1): Market sell subject to MPP. EC MUST handle MPP rejection and retry.

Logger: exit reason = MAE_THRESHOLD_BREACH

CIATS Trade Outcome Bus fires

11.3 Emergency SL (Layer 3 — System Failure Only)

VPS crashes / TothBot dies / internet severs / unrecoverable WS disconnect

Kraken matching engine fires off-book emergSL stop-market at $ATR \times 3.0x$

MPP NOTE: Triggered market order subject to MPP. Assign to 0411001.

On restart: reconciliation detects closed position → Logger: EMERGENCY_SL_FIRED

THIS IS A SYSTEM FAILURE EVENT. SHOULD NEVER FIRE IN NORMAL OPERATION.

12. Open Items — Require Specification Documents

OI-001: Signal generation (SSS) — FP/DP derivation required. Blocks 0811001, 0811002. OI-002: Asset selection universe — Kraken USDT/USDC pairs. OI-003: Exit Controller full specification — MPP handling, cancel timeout fallback, ticker-driven drawdown, batch_cancel for Full Halt [dv1_6]. OI-004: Entry limit placement — evaluate bbo trigger, L2 book (PL-010), L3 book wss://ws-l3.kraken.com/v2 [URL corrected dv1_6]. OI-005: Selection Controller — 6-gate spec, must include per-pair status check [dv1_6]. OI-006: CIATS specification — Mann-Whitney $\alpha=0.01$, Spearman $|\rho|>0.3$, CUSUM k/h, PDCA 50-trade interval [dv1_6]. OI-007: AA=04 Exchange Interface formal document. OI-008: All AA=10 coding specs. WD documents active: 1011001, 1011002, 1011010, 1011012, 1011013. All dv1_2 documents: 0411001 dv1_2, 1011002 dv1_2 (updated dv1_9).

13. Revision History

dv1_0 03-23-2026 Initial document.

dv1_1 03-23-2026 Added all six system flow diagrams.

dv1_2 03-24-2026 Six changes: C-1 emergSL terminology, C-2 rate limits, C-3 OHLC stall,

A-1 ratecounter, A-2 req_id, A-3 latency logging. dv1_3 03-24-2026 Rebuilt as correct .docx with all six diagrams embedded. dv1_4 03-24-2026 Seven changes: C-1 MPP, A-1 AR-001 expansion, A-2 cl_ord_id, A-3 snap_orders, A-4 stp_type, A-5 reconnect distinction, A-6 OTO rejected. dv1_5 03-24-2026 Eleven changes: FLAG1 ohlc(60), FLAG2 REST GetOHLCData, I-1 Gate 3, I-2 cache, I-3 Position Mirror O(1), I-4 Parameter Store snapshot, I-5 Logger async queue, I-6 cancel timeout, I-7 ticker drawdown, I-8

WS dispatch table, I-9 ATR incremental. dv1_6 03-24-2026 Six changes from 10-search FP/DP validation + Operations/CIATS analysis: (1) amend_order primary, edit_order deprecated, batch_cancel, L3 URL corrected. (2) Rate table updated. (3) A-7 WS ping, A-8 zombie detection, A-9 balances seq gap. (4) Per-pair instrument status check, AR-018/019/020. (5) GetOHLCData response[-1] exclusion — CRITICAL correctness requirement. (6) Auth verification, CIATS statistical thresholds. dv1_9 03-25-2026 Six new architectural requirements and five improvements from second 10-search FP/DP organism validation session (TB00010): A-15 (GAP-A CRITICAL) order_status:true mandatory in executions subscription — without it amended/restated silently dropped, breaking A-12/A-13. A-16 (GAP-B) OHLC interval_begin replaces deprecated timestamp. A-17 (GAP-C/D) instrument price_increment replaces tick_size; qty_min/cost_min mandatory Gate 8 validation. A-18 (GAP-E) websockets max_size=10MB + open_timeout=10 mandatory. A-19 (IMP-11) maxratecount from executions ACK is operative rate threshold. IMP-13 subscription warnings[] parsed/logged. IMP-14 amend_id logged. IMP-15 QueryOrdersInfo for targeted reconciliation. AR-026 through AR-031 added. 0411001 updated to dv1_2. 1011002 updated to dv1_2.

dv1_8 03-25-2026 Eight changes from 10-search FP/DP organism validation session

(TB00009 Part 2): A-11 executions cancel_reason deprecated — use reason field. A-12 complete exec_type enum handling required. A-13 restated exec_type handling — engine amend, not fill/cancel. A-14 PL-014 RESOLVED — Spot REST auth confirmed unchanged (Oct 2025 change was Futures REST only). IMP-06 aiohttp mandatory for all REST calls. IMP-07 asyncio TaskManager pattern for 24/7 memory safety. IMP-08 systemd WatchdogSec=120 with 30s ping loop. New AR-022 through AR-025. New WD document 1011013 VPS_Deployment_Coding_Spec dv1_0 created. 0411001 updated to dv1_1. 1011002 updated to dv1_1. Diagrams 1 and 6 updated.

dv1_7 03-25-2026 One change from 10-search FP/DP organism validation session (TB00009):

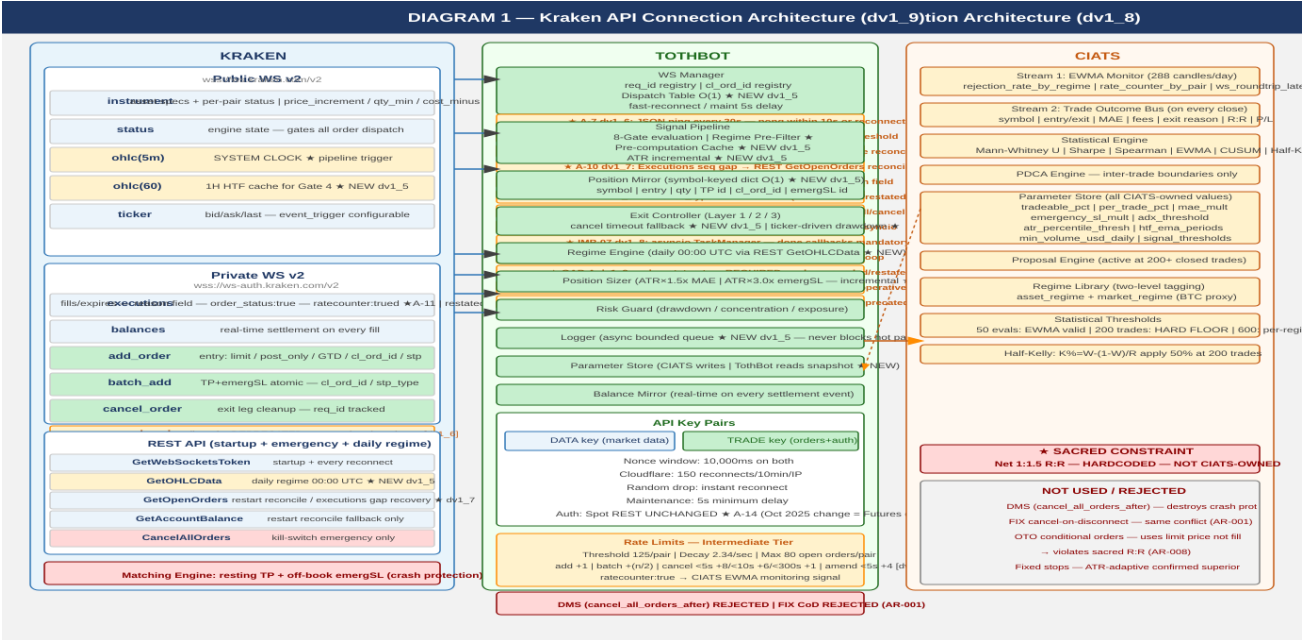
A-10: Executions channel sequence gap detection. Every executions message carries a sequence integer. WS Manager tracks it; any gap triggers REST GetOpenOrders reconciliation immediately. Missed execution events leave Position Mirror in stale state — highest-severity gap class. AR-021 added. GetOpenOrders promoted from restart-only fallback to active gap recovery channel. Diagrams 1 and 6 updated with A-10.

EMBEDDED DIAGRAM SOURCE FILES — SVG (dv1_6)

These are the authoritative SVG source files for all 6 diagrams.
When rebuilding the .docx, Claude reads these SVGs, converts to PNG, and embeds them. Do not edit these manually.

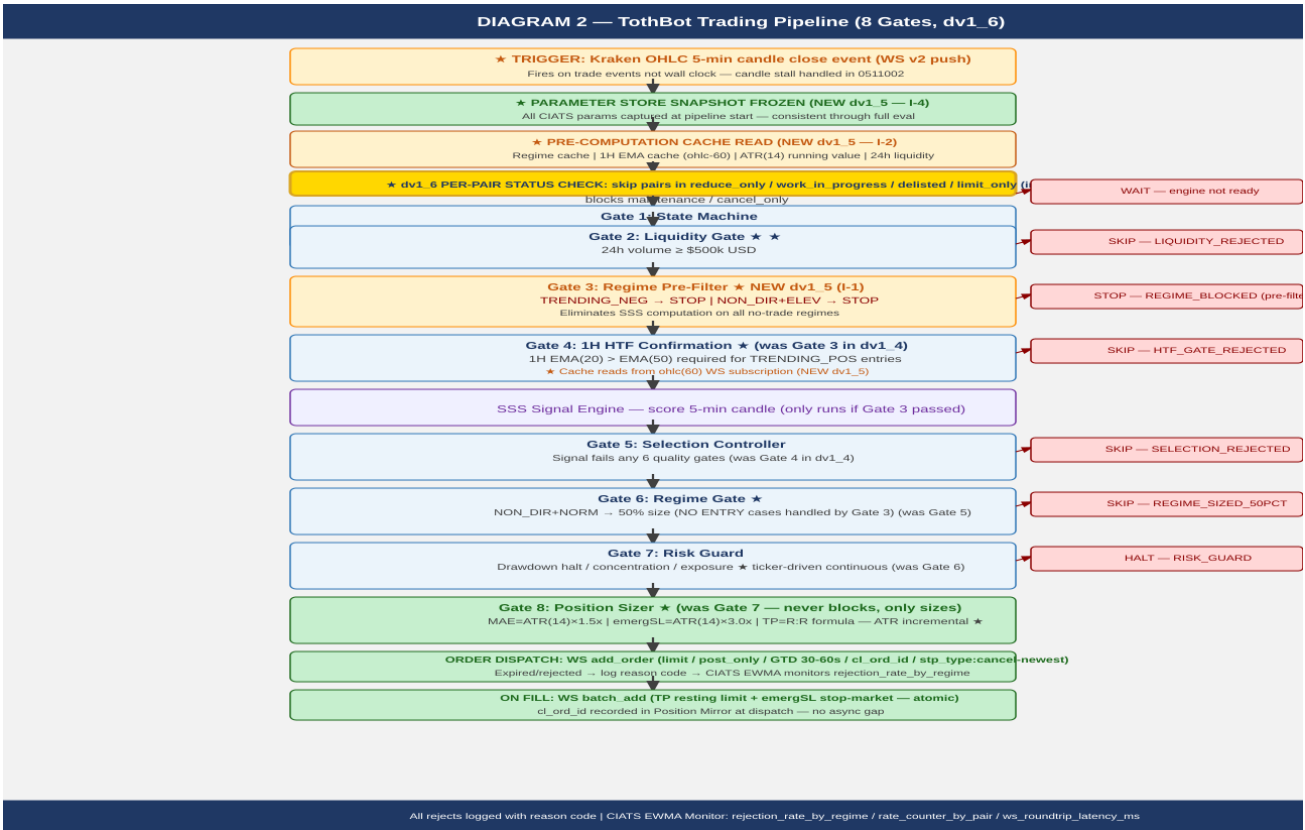
SVG SOURCE: Diagram 1 — Kraken API Connection Architecture

Diagram 1 — Kraken API Connection Architecture (dv1_9)



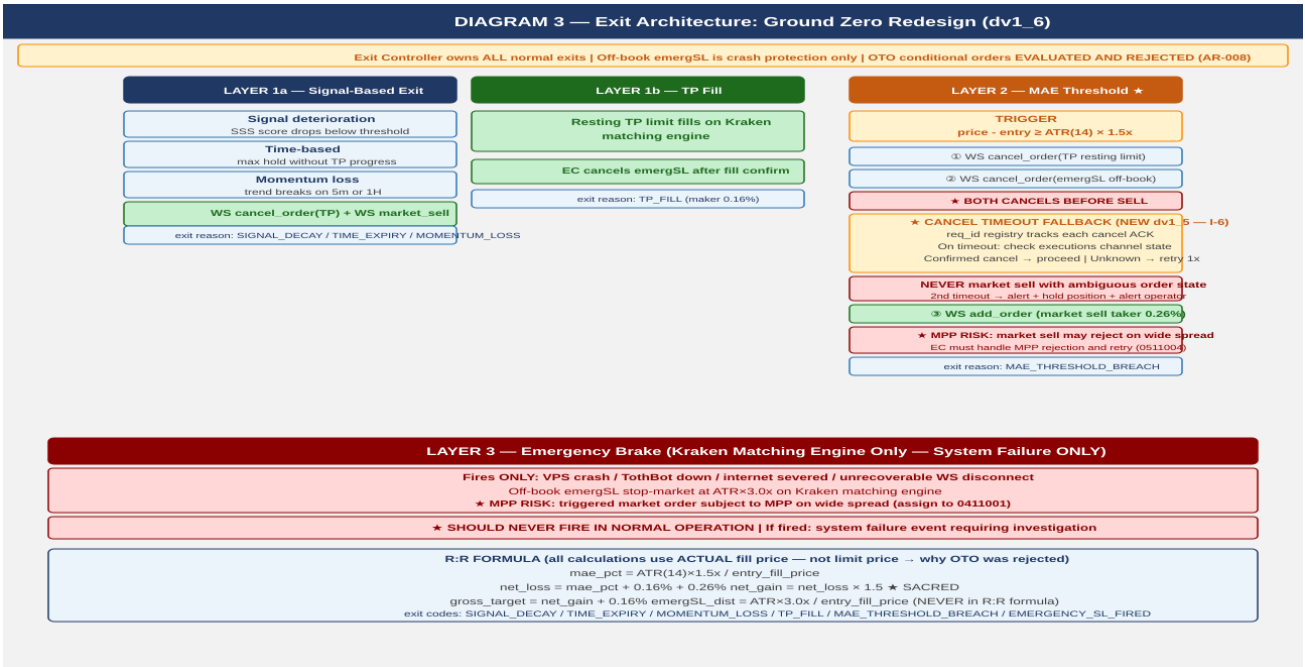
SVG SOURCE: Diagram 2 — TothBot Trading Pipeline (8 Gates)

Diagram 2 — TothBot Trading Pipeline



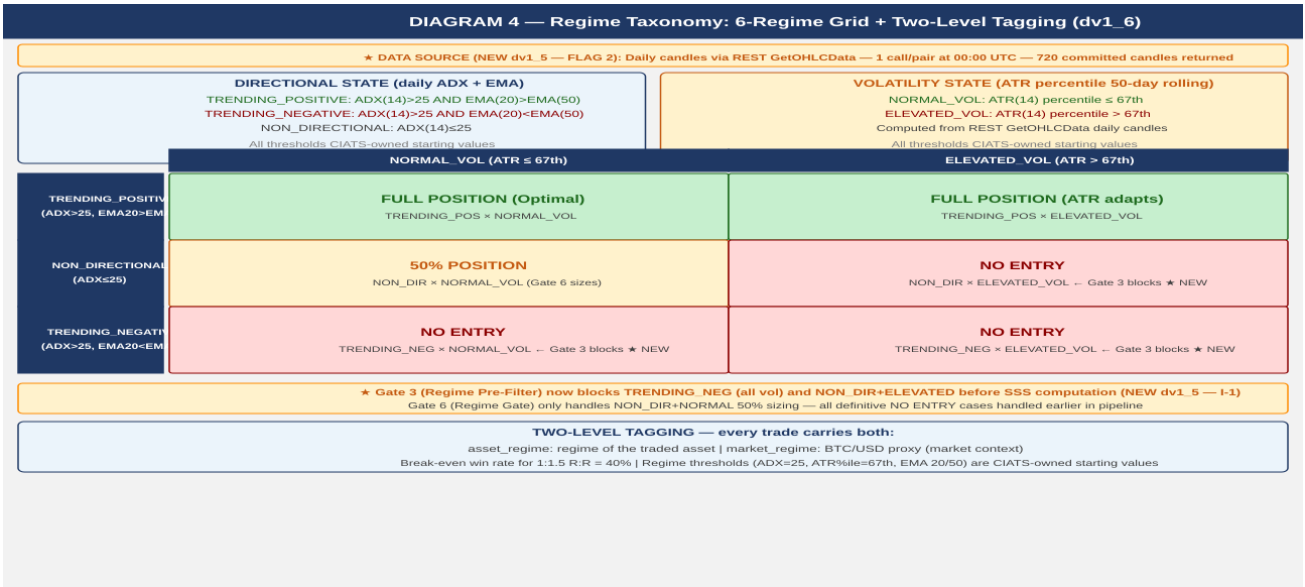
SVG SOURCE: Diagram 3 — Exit Architecture: Ground Zero Redesign

Diagram 3 — Exit Architecture: Ground Zero Redesign



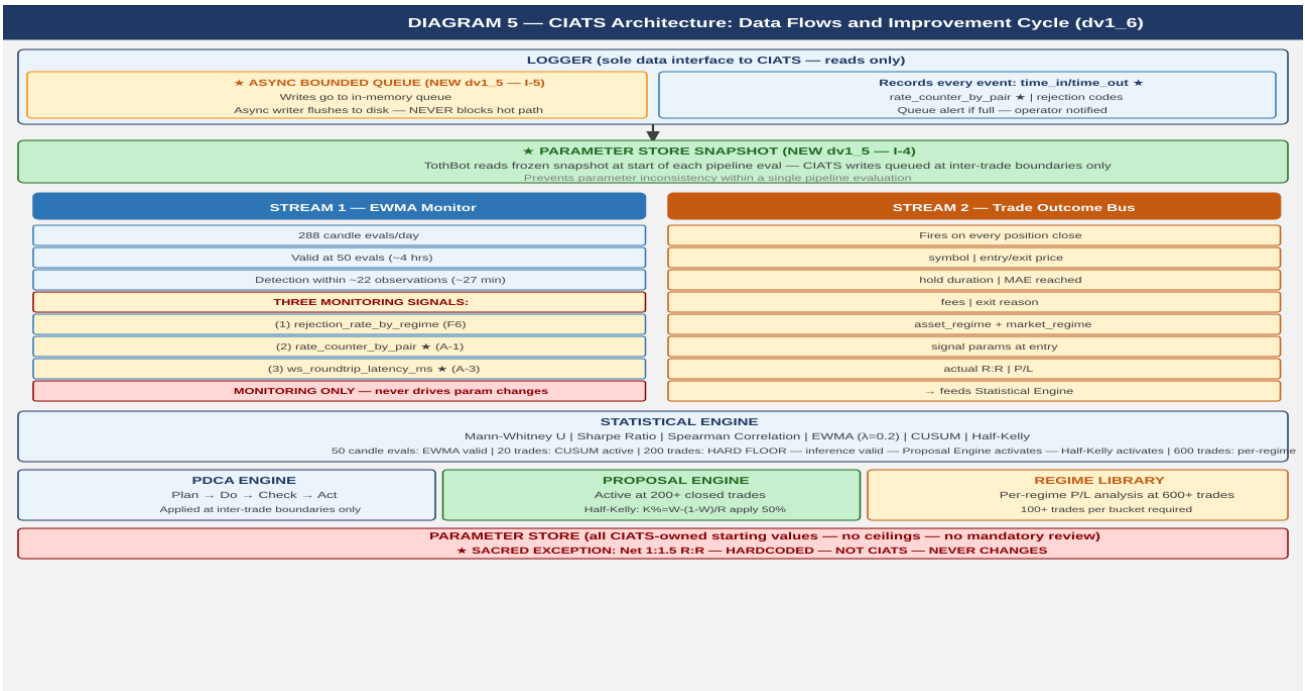
SVG SOURCE: Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging

Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging



SVG SOURCE: Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle

Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle



SVG SOURCE: Diagram 6 — Complete System: One CIATS-Owned Organism

Diagram 6 — Complete System: One CIATS-Owned Organism (dv1_9)

